MIL-C-46113B(MR)
9 February 1973
SUPERSEDING
MIL-C-46113A(MR)
7 June 1968

MILITARY SPECIFICATION

CUTTING-FLUID CONCENTRATES, TRANSPARENT

1. SCOPE

1.1 Scope. This specification covers transparent concentrates which, when diluted with water, form transparent cutting fluids (see 6.1).

1.2 Classification. The concentrates shall be of the following types (see 6.2).

Type I - Consisting primarily of salts, organic surfactants, and other additives

Type II - Consisting primarily of organic surfactants and other additives, and petroleum oil or other lubricity agents.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-T-620	- 1,1,1-Trichloroethane, Technical, Inhibited (Methyl Chloroform)
QQ-A-250/4	- Aluminum Alloy 2024, Plate and Sheet
QQ-B-613	- Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet and Strip)
QQ-C-576	- Copper, Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges (Plate, Bar, Sheet, and Strip)
QQ-I-652	- Iron Castings, Gray
QQ-S-698	- Steel, Sheet and Strip, Low Carbon
UU-T-450	- Tissue, Facial

/FSC 9150/

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM Test Methods and Specifications:

D 92	Flash and Fire Points by Cleveland Open Cup
D 94	Saponification Number of Petroleum Products
D 95	Water in Petroleum Products and Bituminous Materials by Distillation
D 97	Pour Point
D 130	Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test
D 270	Sampling Petroleum and Petroleum Products
D 1193	Reagent Water (D 1193-72 or later revisions)

(The ASTM test methods and specifications listed above are included in Part 17, 18, or 23 of the Annual Book of ASTM Standards and are available individually. Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Specifications and standards of technical societies are generally available for reference from libraries. They are also distributed among using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials. Each concentrate shall be a stable, homogeneous mixture of the constituents specified in 1.2. Each concentrate shall contain sufficient bactericide for satisfactory performance under normal conditions of use.

3.2 Physical and chemical requirements. Each concentrate, its mixtures, and residues, shall conform to the applicable requirements specified in Table I and in 3.3 through 3.10.

Table I. Physical and chemical requirements
(see 4.6, Table III)

Properties	Values	
	Type I	Type II
pH value	9 to 11	9 to 11
Saponifiable oil, wt. percent	0 to 2	5 to 10
Water content, vol. percent, max.	55	45
Pour point, °F, max. (°C, max.)	35 (1.7)	35 (1.7)
Flash point, °F, min. (°C, min.)	Not applicable	300 (149)

3.3 Transparency. The concentrate and its solutions shall be transparent when it is tested in accordance with 4.6.2 for the applicable conditions specified in 3.3.1 through 3.3.4. The fluid shall be considered transparent when the filament of the electric light bulb is discernible through the fluid (see 4.6.2(e)).

3.3.1 The as-received concentrate shall be transparent when it is tested in accordance with steps (a), (d), and (e) of 4.6.2.

3.3.2 A sample of concentrate that has passed the test for accelerated storage stability (see 4.6.3) shall be transparent when it is tested in accordance with steps (a), (d), and (e) of 4.6.2.

3.3.3 A sample of concentrate (diluted with synthetic hard water) that has passed the test for solution stability (see 4.6.4) shall be transparent when it is tested in accordance with steps (b), (d), and (e) of 4.6.2.

3.3.4 A sample of concentrate that has passed the test for miscibility (see 4.6.6) shall be transparent when it is tested in accordance with steps (c), (d), and (e) of 4.6.2.

3.4 Accelerated storage stability. The concentrate shall show no separation, phase-change, gelling, or skinning when it is tested in accordance with 4.6.3.

3.5. Solution stability. The concentrate shall show no separation or phase-change when it is tested in accordance with 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

3.6 Frothing. The volume of froth on the surface of a solution in the graduated cylinder shall not exceed 1.0 ml. when the concentrate is tested in accordance with 4.6.5.

3.7 Miscibility. The residue from the evaporation procedure shall redissolve to form a homogeneous solution, when the concentrate is tested in accordance with 4.6.6

3.8 Corrosion. There shall be no staining, pitting, or etching on brass, cast iron, copper, steel, or aluminum, when the concentrate is tested in accordance with 4.6.7.

3 9 Toxicity. Cutting fluid concentrates furnished under this specification shall have no adverse effect on human health when they are used as intended (see 6.1) Questions concerning product toxicity shall be referred by the procuring activity to the appropriate departmental medical service (for the Army: The Surgeon General, HQDA, ATTN: DASG-DDL-T, Washington, D. C. 20315). Determinations of product toxicity shall be made in accordance with 4.6.8.

3.10 Workmanship Each concentrate shall be a clear, stable, homogeneous fluid, free from objectionable odors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot

4.2.1 Bulk lot. An indefinite quantity of a homogeneous mixture of one type of concentrate offered for acceptance in a single isolated container; or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of 55-gallon drums or smaller unit containers of identical size and type, offered for acceptance at one time, and filled with a homogeneous mixture of one type of concentrate from a single isolated container; or filled with a homogeneous mixture of one type of concentrate manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment with no change in the ingredient materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

4.3 Sampling.

4.3.1 Sampling for the examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at Inspection Level II and Acceptable Quality (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM Method D 270.

4.4 Inspection. Perform inspection in accordance with Method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of filled containers. Examine samples taken in accordance with 4.3.1 for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests. All tests are quality conformance tests.

4.6 Tests. Perform tests in accordance with Table II and with 4.6.1 through 4.6.8, as applicable. References to reagent-grade water shall be understood to mean water conforming to Type IV of ASTM Specification D 1193-72 (or later revisions) for Reagent Water. Water of higher purity (Type II or III) may be used if more convenient, however.

Table II. Test methods

Test	ASTM Method No.
Saponifiable oil	D 941
Water content	D 95
Pour point	D 97
Flash point	D 92

[1]In calculating the percent of saponifiable oil, the saponification number of the fatty oil shall be taken as 195.

4.6.1 pH value. Using any recognized standard make of pH apparatus, determine the pH value of the concentrate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

4.6.2 Transparency.

(a) When performing the test on either the as-received concentrate or on the sample from the accelerated storage-stability test (see 4.6 3), dilute 20 ml. of the concentrate with 400 ml. of reagent-grade water (see 4.6). Fill a 250 ml. beaker with this mixture to a height of 3 ± 1/8 inches (75 ± 3 mm.)

(b) When performing the test on the sample from the solution stability test (see 4.6.4), fill the pour point test jar with the solution to a height of 3 ± 1/8 inches (75 ± 3 mm.)

(c) When performing the test on the re-dissolved residue from the miscibility test (see 4.6.6), dilute 10 ml. of the re-dissolved residue with 150 ml. of reagent-grade water (see 4.6) and fill the pour point test jar with this mixture to a height of 3 ± 1/8 inches (75 ± 3 mm).

(d) Place a lighted, unfrosted, 7-watt, 110-volt electric light bulb against the bottom of the beaker.

(e) Examine the light bulb through the cutting-fluid solution from the top of the beaker. (The cutting-fluid solution shall be considered transparent if the filament of the light bulb is discernible.)

4.6.3 Accelerated storage stability. Pour a 50-ml. sample of the concentrate into a glass-stoppered, 100-ml. graduated cylinder. Place the cylinder in an oven maintained at 160° ± 5°F (71° ± 3°C) for 5 hours. Remove the cylinder from the oven and allow the sample to cool to 75° ± 5°F (24° ± 3°C). Place the cylinder in a refrigerator maintained at 10° ± 2°F (-12° ± 1°C) for 24 hours. Remove the cylinder from the refrigerator. Allow the sample to return to 75° ± 5°F (24° ± 3°C) and to stand at this temperature for one hour without agitation. Examine the sample at the end of the standing period for conformance with 3.4. If none of these conditions is found, test the sample for conformance to the transparency requirement of 3.3.2.

4.6.4 Solution stability. Prepare a solution of synthetic hard water by dissolving 0.400 grams of ACS Reagent Grade anhydrous calcium chloride in one liter of reagent-grade water (see 4.6). Pour 95 ml. of this solution into a glass-stoppered, 100- ml. graduated cylinder. Cool the solution to 45° ± 2°F (7.2° ± 1°C). Add 5 ml. of the concentrate, stopper the cylinder, and shake the mixture vigorously for one minute. Allow the mixture to stand for 25 hours at 75° ± 5°F (24° ± 3°C) and examine it for separation of phase-change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

4.6.5 Frothing. Pour 190 ml. of reagent-grade water (see 4.6) into a glass-stoppered, 250-ml. graduated cylinder. Add 10 ml. of the concentrate and shake the mixture vigorously for one minute. Allow the mixture to stand for 15 minutes at 75° ± 5°F. (24° ± 3°C.) and observe the volume of froth remaining above the liquid.

4.6.6 Miscibility. Pour 50 ml. of the concentrate into a low-form, 1000-ml. beaker. Place the uncovered beaker in a forced-draft oven maintained at 155° to 165°F (68° to 74°C) for 16 hours. Remove the beaker from the oven at the end of this period and allow the sample to cool to 75° ± 5°F (24° ± 3°C). Add 40 ml. of reagent-grade water (see 6.3), in 20-ml. increments, to the residue in the beaker. Stir the contents thoroughly with a glass stirring rod after each addition of water. Transfer the contents of the beaker to a 100-ml. graduated cylinder. Fill the cylinder to the 100-ml. mark with reagent-grade water and allow the solution to stand for one hour. Examine the solution for homogeneity and test it for conformance to the transparency requirement of 3.3.4.

4.6.7 Corrosion.

4.6.7.1 Equipment. The following items of equipment are required for the corrosion test:

A saucer or plate: glass, pottery or china
Five test tubes, approximately 32 mm. in O.D. by 2(
in length
A pair of stainless steel forceps
A magnifier of 5X power

4.6.7.2 Materials. The following materials are required for the corrosion test:

4.6.7.2.1 Metal test specimens. Except for the cast iron chips, all metal test specimens shall be cut from strip stock and shall measure approximately 3 by 1/2 by 1/8 inches (76 by 13 by 3 millimeters).

Twenty-eight grams of clean, dry, cut cast-iron chips
Two aluminum specimens conforming to QQ-A-250/4, temper F
Two brass specimens conforming to QQ-B-613, composition 2
(cartridge brass), half-hard temper
Two copper specimens conforming to QQ-C-576, electrolytic
tough-pitch, cold-rolled, half-hard temper
Two cast-iron specimens conforming to QQ-I-652, class 20
Two steel specimens conforming to QQ-S-698, physical
quality cold-rolled, half-hard temper (or half-hard
No. 2 temper)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

4.6.7.2.2 Solvents, chemicals, and miscellaneous materials.

1,1,1-trichloroethane (methyl chloroform) conforming to Q-T-620.
Reagent-grade water (see 4.6).
Tissue paper conforming to UU-T-450.

4.6.7.3 Procedure. Prepare a mixture of the cutting fluid by pouring 285 ml. of reagent-grade water and 15 ml of the concentrate into a flask of sufficient size. Stopper the flask and invert it several times to mix the contents thoroughly.

4.6.7.3.1 Testing of the metal-strip specimens. Prepare the metal strip specimens described in 4.6.7.2.1 (but not the cast iron chips) in accordance with ASTM Method D 130 Using forceps, place one specimen of each of the five different metals in a separate test tube. Place all of the five duplicate specimens in a bottle filled with 1,1,1-trichloroethane. Keep this bottle tightly stoppered until ready to examine the specimens.

WARNING

1,1,1-trichloroethane is a toxic solvent. Avoid breathing the vapor. Do not allow the liquid to contact eyes, skin, or clothing. Use with adequate ventilation.

Pour 50 ml. of the cutting fluid mixture into each of the five test tubes containing one each of the five metal-strip specimens to be tested. Set the test tubes aside for 48 hours at 75° ± 5°F (24° ± 3°C). At the end of this test period, remove the specimens from the test tubes Wash the specimens in 1,1,1-trichloroethane and allow them to dry in air. Remove the five duplicate specimens from the bottle of 1,1,1-trichloroethane and allow them to dry in air. Examine and compare the exposed and duplicate specimens under 5X magnification to determine whether staining, pitting, or etching has occured on the exposed specimens.

4.6.7.3.2 Testing of the cast-iron chip specimens. Pour the remaining 50 ml. of the fluid mixture specified in 4.6.7 3 into a 100 ml beaker. Add 28 grams of dry, cut cast-iron chips. Stir the mass with a glass rod for 30 seconds. Drain off the fluid mixture and dump the wet chips in a pile on the saucer. Allow the chips to dry undisturbed for 24 hours at 75° ± 5°F (24° ± 3°C). Examine the dried chips for evidence of rust.

4.6.8 Toxicity. The supplier shall furnish the procuring activity with all information necessary to evaluate the safety of the product. Such information shall be furnished in one of the following forms, at the discretion of the procuring activity:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

a. A complete listing of the constituent materials, giving the percentages of composition and using standard chemical nomenclature. (This is the simplest and most direct method. In most instances it will completely satisfy the requirement for information.)

b. Results of toxicological testing of a scope and quality acceptable to the procuring activity.

c. Some other form acceptable to the supplier and to the procuring activity.

All information furnished by suppliers in connection with the evaluation of toxicity shall be signed by a responsible official of the supplying firm. Such information shall be held in strict confidence by the procuring activity and shall not be divulged to other suppliers.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Unless otherwise specified in the contract or order (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290. In addition, the following cautionary note shall be placed on each container:

"CAUTION! DO NOT STORE BELOW 40°F (4°C)"

6. NOTES

6.1 Intended use. Concentrates covered by this specification are intended for use in cutting fluids in machines and grinding operations. One part of concentrate can be diluted with 20 to 80 parts of water, by volume.

6.1.1 Type I concentrate is intended for use when the major considerations are wetting, cooling, and free cutting.

6.1.2 Type II concentrate is intended for use when the major consideration is surface finish.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of concentrate (see 1.2).
(c) Quantity of concentrate.
(d) Type and size of container (see 5.1).
(e) Level of packaging and packing (see 5.1).

Custodian:
Army - MR

Preparing activity:
Army - MR

Review Activities:
Army - MU, WC, MD
DSA - PS

Project No. 9150-A216

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity

SPECIFICATION

MIL-C-46113B(MR) Cutting-Fluid Concentrates, Transparent

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
Army Materials and Mechanics Research Center
Watertown, Massachusetts 02172

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Director
Army Materials and Mechanics Research Center
ATTN: AMXMR-MS
Watertown, Massachusetts 02172

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:54Z
Check the source to verify that this is the current version before use.